



The Foreign Exchange Market

How currencies are traded, quoted, and exchanged in the world's largest financial market

Roadmap: What We Cover in Chapter 5

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FX Market Functions

Three core purposes: transfer purchasing power, provide credit, manage currency risk

02

Market Structure & Players

Five participant types, 24-hour global trading, \$5.1 trillion daily volume

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Evolution of FX Trading

From phone calls in the 1980s to electronic platforms and algorithmic trading today

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Direct vs. indirect, bid/ask spreads, cross rates, triangular arbitrage

01

FX Market Functions

Why does the foreign exchange market exist
and who needs it?

Three Core Functions of the Foreign Exchange Market

The FX market is the mechanism that allows any economic actor — from a multinational firm to an individual tourist — to convert one currency into another.

1. Transfer Purchasing Power



Enables economic actors to move buying power across currency borders.

Examples:

- Vietnamese electronics firm receives USD from a U.S. buyer; converts to VND to pay workers
- A U.S. tourist exchanges dollars for Thai baht before visiting Bangkok
- Apple converts EUR revenues from Europe back into USD for its U.S. balance sheet

2. Provide Credit for Trade



International trade often requires financing during shipment time (30–90 days). Banks and FX markets help exporters and importers trade today even though payment and delivery happen at different times.

Examples:

- A Brazilian coffee exporter ships to a German buyer; a bank provides a letter of credit bridging the time gap
- An importer needs 60 days to sell goods before paying the supplier in foreign currency
- Banks use FX swaps to fund cross-border trade financing

3. Minimize Exchange Rate Risk



Businesses hedge currency exposure to protect profit margins.

Examples:

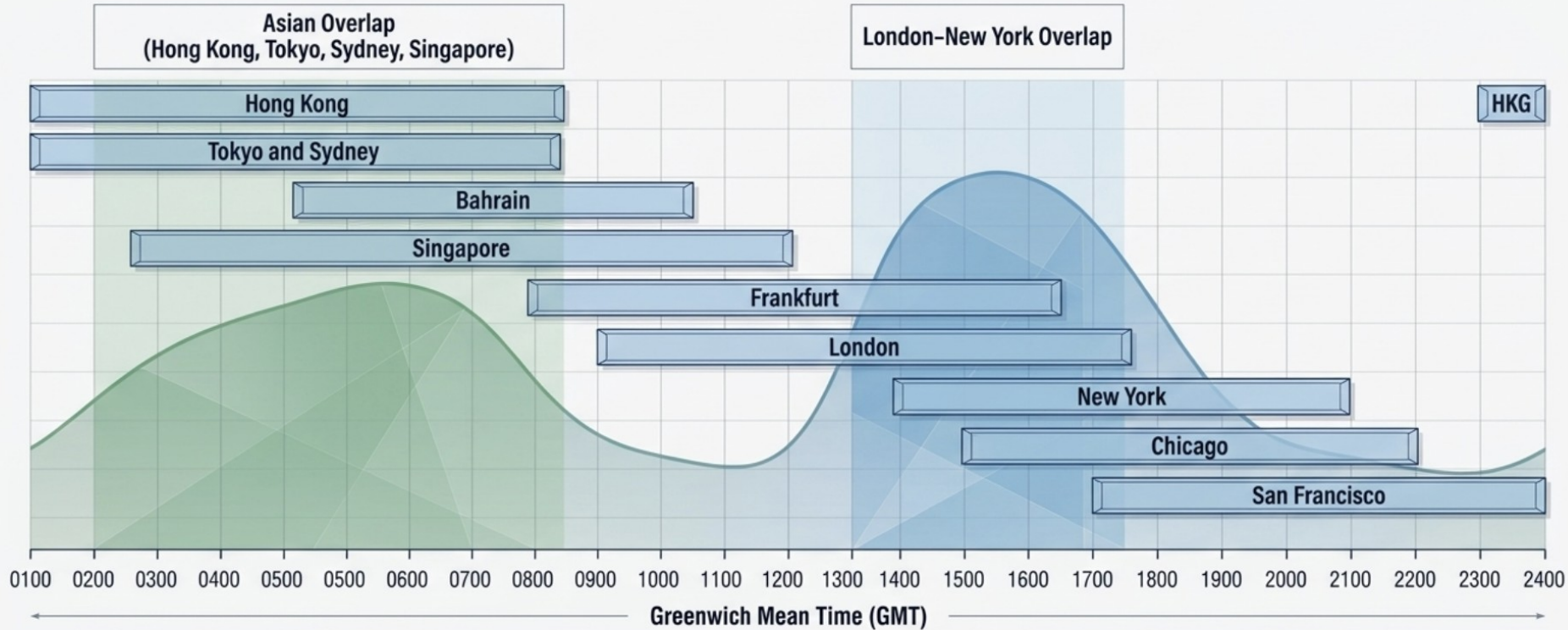
- Airbus locks in a USD/EUR forward rate when signing a USD-denominated jet sale contract
- A Japanese automaker sells cars in the U.S. but pays costs in JPY — buys JPY forward to hedge
- A Vietnamese manufacturer exporting to the U.S. buys USD forward to protect its VND revenue

02

Market Structure & Players

Who trades in the FX market and how is it organized?

The Continuous Cycle of Global Liquidity



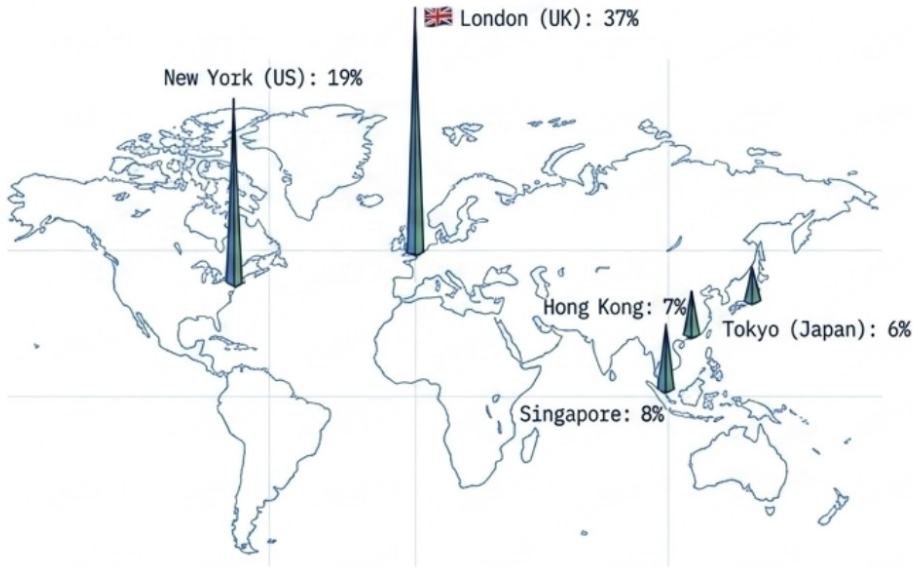
The FX market operates 24 hours a day, shifting westward across the globe.

Peak depth and liquidity occur during major geographic overlaps. Historically dominated by the London/New York window, activity is increasingly shifting East during the Asian trading overlap.

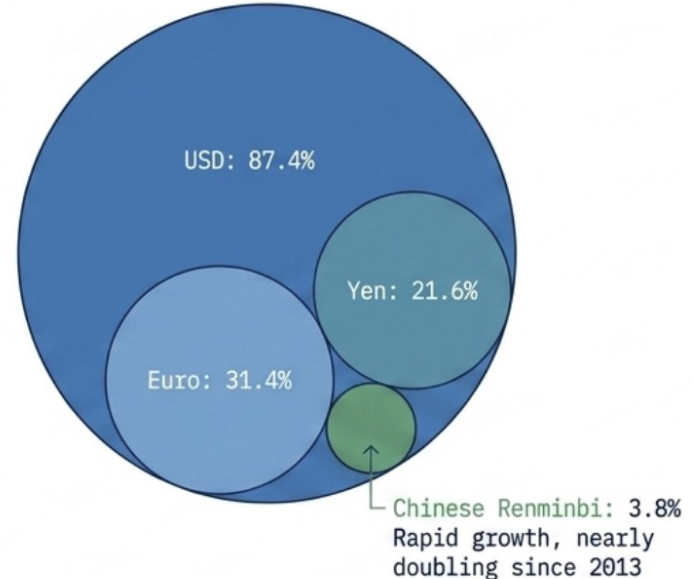
The Uncontested Centers of Gravity

Despite a highly decentralized global network, trading volume clusters intensely around the UK for execution and the US Dollar as the universal base mechanism.

Geography



Currency



Five Types of FX Market Participants

Every participant has a different motivation — from earning the bid/ask spread to protecting a nation's currency value.



Bank & Nonbank Dealers

Goal: Profit from bid/ask spread

JPMorgan quotes EUR/USD 1.2170 bid / 1.2178 ask. It buys euros cheaply, sells at a higher price — pocketing the 0.0008 spread.



Commercial & Investment Transactors

Goal: Facilitate underlying business

Toyota sells cars in the U.S., receives USD, must convert to JPY to pay Japanese workers and suppliers.



Speculators & Arbitrageurs

Goal: Profit purely from currency movements

A hedge fund takes a \$500M long position on GBP, betting it will rise after a Bank of England rate decision.



Central Banks & Treasuries

Goal: Manage national currency policy

Bank of Japan sells JPY and buys USD when the yen becomes 'too strong,' which hurts Japanese exporters.



FX Brokers

Goal: Earn a commission, not FX profit

An electronic broker like EBS matches a bank wanting to sell EUR with another bank wanting to buy EUR.

The Ecosystem Matrix: Market Participants

	Primary Motive	Market Role	Risk Profile
Bank & Nonbank Dealers	Bid/Ask Spread Profit	Market Makers (maintaining inventory)	Moderate (Liquidity providers)
Commercial & Investment	Utility (incidental to underlying business)	Market Takers	Low (Focus on execution)
Speculators & Arbitragers	Absolute Profit (exploiting price/rate changes)	Opportunists	High (trading for their own interest)
Central Banks & Treasuries	Policy / FX Intervention	Market Influencers	Willing Loss Takers (motive is citizen benefit, not profit)
Foreign Exchange Brokers	Commission	Facilitating Agents	Zero (Do not hold principal positions)

03

Evolution of FX Trading

From phone calls and paper tickets to
electronic platforms and algorithms

How FX Trading Evolved: Three Eras

1970s–1980s

The Telephone Era

- All trades done by phone — voice quotes, handwritten tickets
- Traders didn't know what rates competitors were quoting
- Settlement was paper-based and error-prone
- Reuters launched FAFX page (1987) — first electronic price display
- Key risk: Herstatt Bank (1974) collapse showed settlement risk

1990s–2000s

The Electronic Era

- Electronic brokers (EBS, Reuters) replace voice brokers
- Traders see real-time prices; know counterparty identity
- State Street launches FXConnect (1996) for retail aggregation
- CLS Bank opens (2002) to solve Herstatt-style settlement risk
- Interdealer and customer markets still largely separate

2010–Present

The Algorithm Era

- Multibank (MBT) and single-bank (SBT) electronic platforms collapse old barriers
- Retail customers access institutional pricing via aggregators
- 75%+ of trades executed electronically (2014)
- Algorithmic and high-frequency trading dominate volume
- 2013–14 fixing scandal: traders colluded via chat rooms to move the 4pm 'London fix'

FX Market Manipulation: The London Fix Scandal (2013–2014)

The 'London Fix' is the 4:00 PM daily benchmark rate used by pension funds, index trackers, and corporations worldwide to value their holdings.

What Went Wrong

→ Traders at major banks shared client order flows in private chat rooms ('The Cartel', 'The Mafia')

→ They coordinated to move rates at 4pm, profiting at clients' expense

→ Fines: Barclays, Citi, JPMorgan, UBS and others paid \$10B+ in penalties

→ Similar to LIBOR scandal — exposed human element in price-setting

The Electronic Fix

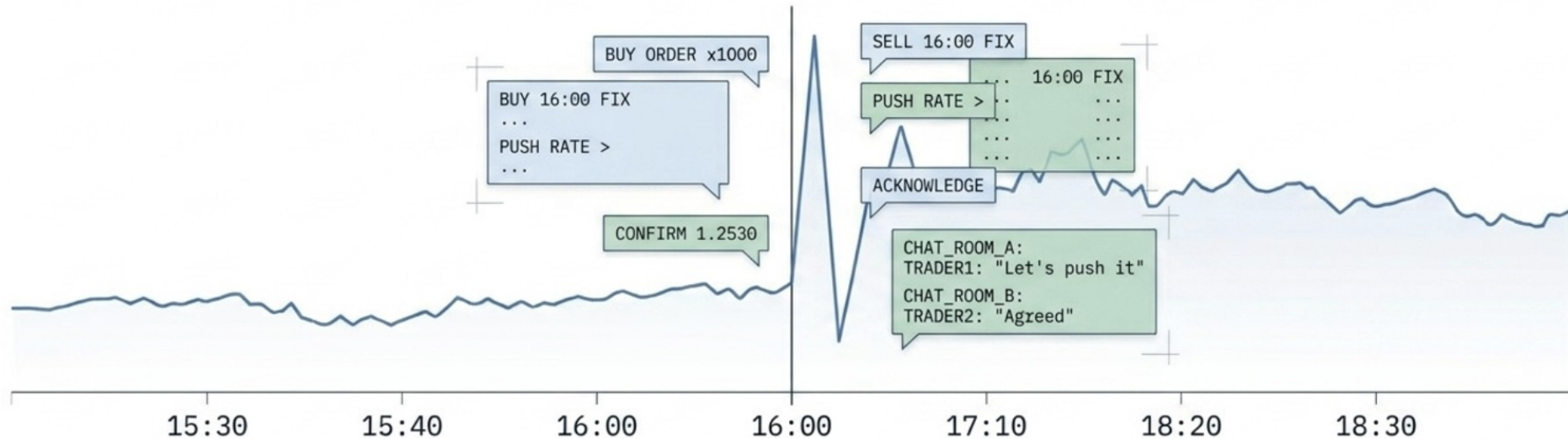
→ 75% of all trades executed electronically by 2014 — harder to collude

→ Algorithmic trading removes human discretion at key fix times

→ But: sophisticated algorithms can still manipulate in subtler ways

→ Lesson: technology changes the method of manipulation, not the incentive

The Persistent Human Element: Fixing the Fix



The Issue

Following the LIBOR scandals, scrutiny fell on the London fix—the 4 p.m. daily benchmark rate. Traders colluded via chat rooms and phone calls to manipulate prices at key benchmark times.

The Reality

By 2014, nearly 75% of trades were electronic. While this was expected to fix market spikes caused by collusion, electronic trading merely facilitated market manipulation of a more sophisticated kind.

04

Transaction Types

Spot, forward, and swap — the three ways to trade currency

Three Types of FX Transactions: Spot, Forward, Swap

Defined by WHEN delivery and payment actually occur — not when the deal is agreed.

Spot Transaction

Settlement: T+2 (2 business days after deal)

- Deal agreed today at current market rate
- Delivery and payment happen in 2 business days ('value date')
- Most common transaction type; sets the reference price

Example:

Today (Monday): A U.S. importer agrees to buy €1M at EUR/USD 1.2170.
Wednesday: Importer delivers \$1,217,000 and receives €1,000,000.

Forward (Outright Forward)

Settlement: Agreed future date (1 week to 5+ years)

- Rate fixed NOW; payment/delivery at a future 'value date'
- Quoted as spot rate $\pm$ forward points (pips)
- Buying and selling forward are the same trade — just described from different sides

Example:

Airbus signs a contract to sell a jet to Delta for \$200M, delivery in 6 months.
Airbus buys USD 6-month forward at 1.0950 to guarantee its EUR revenue regardless of EUR/USD moves.

Swap Transaction

Settlement: Two dates simultaneously

- SIMULTANEOUSLY buy and sell the same currency for two different dates
- Both legs with the same counterparty
- Types: spot-against-forward, forward-forward, NDF (non-deliverable forward)

Example:

A U.S. bank needs EUR for 3 months but has USD. It sells USD/buys EUR spot today and simultaneously buys USD/sells EUR 3 months forward — effectively 'borrowing' EUR for 3 months.

05

Quotations & Rates

How currencies are priced: direct/indirect, bid/ask, cross rates, and arbitrage

Anatomy of a Currency Quote: Reading EUR/USD 1.2170

Every FX quote has two parts: a *BASE* currency and a *PRICE* (quote) currency. The number tells you how many units of the price currency buy one unit of the base.

$$\text{EUR} / \text{USD} = 1.2170$$

Base currency
(what you buy/sell)

Price currency
(what you pay in)

"1 euro costs
\$1.2170"

American Terms

USD price of 1 foreign currency unit

- EUR/USD 1.2170 → 1 euro costs \$1.2170
- GBP/USD 1.5552 → 1 pound costs \$1.5552
- Used for: euro, pound, AUD, NZD and most options/futures



Think: 'How many dollars does 1 euro cost?'

European Terms

Foreign currency price of 1 USD

- USD/JPY 83.16 → 1 dollar buys 83.16 yen
- USD/CHF 0.9748 → 1 dollar buys 0.97 Swiss francs
- Most common convention globally — majority of pairs quoted this way



Think: 'How many yen does 1 dollar cost?'

Direct vs. Indirect Quotes & the Bid/Ask Spread

These terms describe the SAME rate but from different perspectives. 'Home country' determines which label applies.

Direct vs. Indirect Quote

Direct quote: home currency price of 1 foreign unit

→ For a U.S. resident: EUR/USD 1.2170 (dollars per euro) is direct

Indirect quote: foreign currency price of 1 home unit

→ For a U.S. resident: USD/EUR 0.8214 (euros per dollar) is indirect

Key insight: direct and indirect are RECIPROCALs of each other

Bid / Ask (Offer) Spread

BID = dealer BUYS the base currency (lower price)

→ EUR/USD 1.2170 bid: dealer pays \$1.2170 for €1

ASK = dealer SELLS the base currency (higher price)

→ EUR/USD 1.2178 ask: you pay \$1.2178 to buy €1

Spread (0.0008) = dealer's profit on every unit traded

Real Market Quotes (EUR/USD): How to Read Them

Currency Pair	ISO Code	Bid	Ask	Spread (pips)	Meaning
Euro	EUR/USD	1.2170	1.2178	8	\$1.2170–\$1.2178 per €1
Japanese Yen	USD/JPY	83.16	83.22	60	¥83.16–¥83.22 per \$1
British Pound	GBP/USD	1.5548	1.5556	8	\$1.5548–\$1.5556 per £1
Canadian Dollar	USD/CAD	0.9928	0.9936	8	C\$0.9928–C\$0.9936 per \$1

Cross Rates & Triangular Arbitrage: Catching Mispriced Currencies

A cross rate is the exchange rate between two non-USD currencies, calculated through their USD rates. Discrepancies create arbitrage opportunities.

Given quotes:

Citibank: EUR/USD = 1.3297 (1 EUR = \$1.3297)

Barclays: GBP/USD = 1.5585 (1 GBP = \$1.5585)

Dresdner: GBP/EUR = 1.1722 (1 GBP = €1.1722)

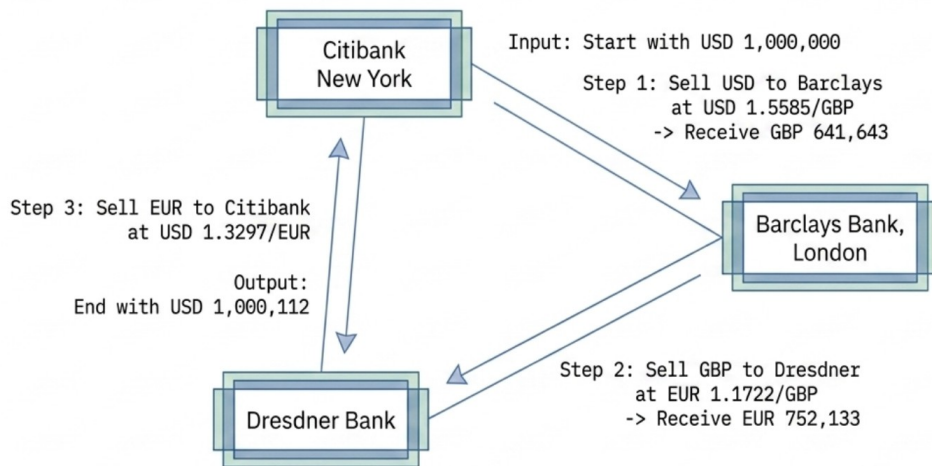
Implied cross rate:

$\text{GBP/EUR} = \$1.5585 \div \$1.3297 = 1.1721$

Dresdner quotes 1.1722 — 0.0001 HIGHER than implied.

This tiny gap = arbitrage profit opportunity!

Exploiting the Cross Rate: The Arbitrage Circuit



Takeaway: Because cross rates occasionally fall out of alignment, a \$112 risk-free profit is extracted instantly, which forces the quotes back into equilibrium.

Forward Quotations: Pips, Points & How to Read Them

Forward rates are quoted as points (pips) added to or subtracted from the spot rate — not as full outright rates.

Forward Rate = Spot Rate + Forward Points (pips / 10,000)

Example: EUR/USD spot bid 1.0897, 3-month forward bid points = +53 → forward rate = 1.0897 + 0.0053 = 1.0950

Positive points = forward at PREMIUM to spot. Negative points (like JPY) = forward at DISCOUNT.

EUR/USD — Forward at Premium (EUR stronger over time)

Term	Bid Points	Bid Rate	Ask Points	Ask Rate
Spot	—	1.0897	—	1.0901
1 month	+17	1.0914	+19	1.0920
3 months	+53	1.0950	+54	1.0955
6 months	+112	1.1009	+113	1.1014
1 year	+242	1.1139	+245	1.1146

EUR trades at a **FORWARD PREMIUM** vs USD → the market expects EUR to appreciate (or US interest rates are higher than EU rates).

USD/JPY — Forward at Discount (JPY stronger over time)

Term	Bid Points	Bid Rate	Ask Points	Ask Rate
Spot	—	118.27	—	118.37
1 month	-51	117.76	-50	117.87
3 months	-143	116.84	-140	116.97
6 months	-288	115.39	-287	115.50
1 year	-584	112.43	-581	112.56

USD trades at a **FORWARD DISCOUNT** vs JPY → market expects JPY to strengthen (Japan's rates near zero → interest rate parity at work).

Chapter 5

Key Takeaways

The Foreign Exchange Market

1

\$5.1 Trillion Daily — No Central Exchange

FX is decentralized, OTC, and 24-hour. London (37%) and NY (19%) dominate but Asia is growing fast.

2

Five Player Types, Five Different Motives

Dealers earn spreads; transactors convert for business; speculators bet on rates; central banks intervene; brokers connect.

3

Three Transaction Types by Delivery

Spot (T+2) for immediate needs, Forward for locking in future rates, Swap to borrow one currency against another.

4

Base/Quote, Bid/Ask — Learn the Jargon

EUR/USD 1.2170/78: dealer buys €1 at \$1.2170 (bid), sells at \$1.2178 (ask). Spread is the dealer's profit.

5

Cross Rates Enable Triangular Arbitrage

Inconsistent quotes across three banks let traders pocket risk-free profits — and in doing so, push quotes back into alignment.